

P232T8 $\langle L, \leq \rangle$ 是格.

$\forall a \in L$ 令 $S = \{x \mid x \in L \wedge x \leq a\}$. 证 $\langle S, \leq \rangle$ 是 L 子格

任取 $x, y \in S$. 则有 $x \leq a$ 且 $y \leq a$

$$x \wedge y \leq x \leq a \quad x \vee y \leq a \vee a \leq a$$

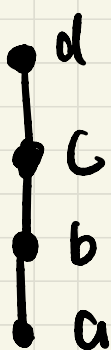
因此 S 对 $\wedge$ 和 $\vee$ 封闭

又 $\because a \in L$ 且 $a \leq a$ 故 $a \in S$

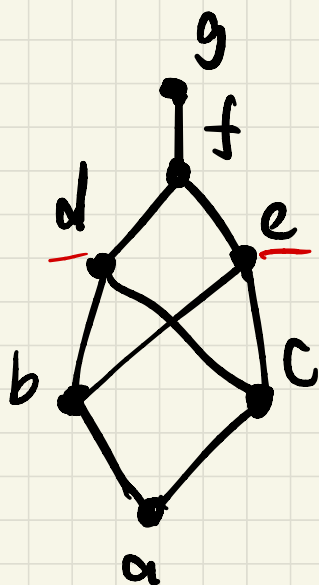
$\therefore S$ 非空

综上所述 $\langle S, \leq \rangle$ 是 L 子格

图 11-11

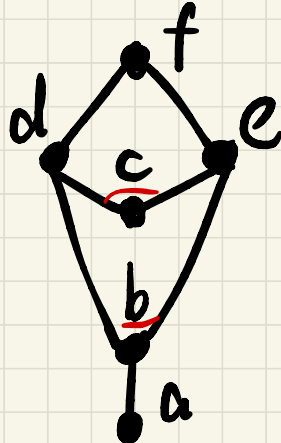


(a) ✓



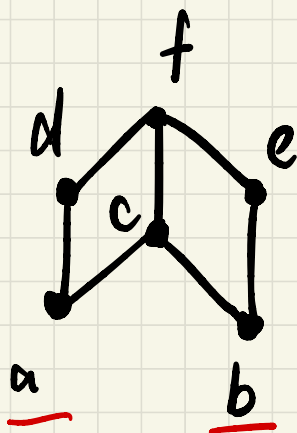
(d) ✗

de 无最大下界



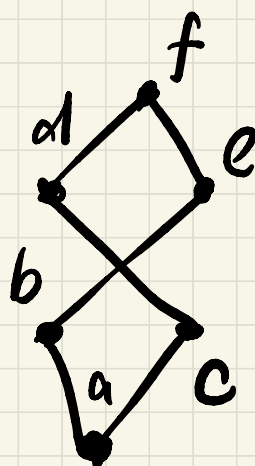
(b)

{d, e} 无最大下界

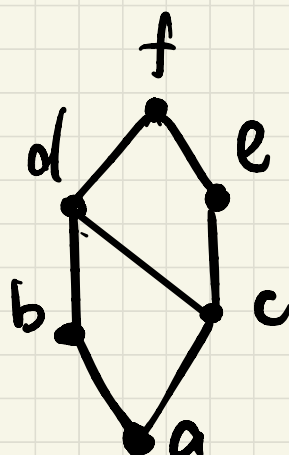


(e) ✗

{a, b} 无最大下界



(c) ✓



(f) ✓

T9. 针对上图求存在补元元集的补元

(a). a 与 d 互为补元. 其他元素不存在

(b) a 与 f 互为补元. b 的补元是 c 和 d

c --- 是 d 和 e

e --- 是 d 和 c

d --- 是 b 和 e

(f) a 与 f 互为补元 b 与 e 互为补元 c 与 d 互为补元

T13 $f = (a \wedge b) \vee (a \wedge b \wedge c) \vee (b \wedge c)$

$$\begin{aligned} (1) f &= (a \wedge b) \vee ((a \wedge b) \wedge c) \vee (b \wedge c) \\ &= (a \wedge b) \vee (b \wedge c) \end{aligned}$$

$$= b \wedge (a \vee c)$$

$$(2) f^* = b \vee (a \wedge c)$$

110 设 $\langle B, \wedge, \vee, ', 0, 1 \rangle$ 是布尔代数

B 上定义 $\oplus$. $\forall x, y \in B$ 有 $x \oplus y = (x \wedge y') \vee (x' \wedge y)$

$\langle B, \oplus \rangle$ 能否构成代数系统?

$$\begin{aligned} x \oplus x &= (x \wedge x') \vee (x' \wedge x) \\ &= 0 \vee 0 = 0 \end{aligned} \quad x \text{ 本身是逆元}$$

$$\begin{aligned} x \oplus 0 &= (x \wedge 1) \vee (x' \wedge 0) \\ &= x \vee 0 = x \end{aligned} \quad 0 \text{ 是单位元}$$

$$\begin{aligned} (x \oplus y) \oplus z &= ((x \wedge y') \vee (x' \wedge y)) \oplus z \\ &= (((x \wedge y') \vee (x' \wedge y)) \wedge z') \vee (((x \wedge y') \vee (x' \wedge y))' \wedge z) \end{aligned}$$

$$= (x \wedge y' \wedge z') \vee (x' \wedge y \wedge z') \vee ((x \vee y) \wedge (x \vee y') \wedge z)$$

$$x \oplus (y \oplus z) = x \oplus ((y \wedge z') \vee (y' \wedge z))$$

$$= (x \wedge ((y \wedge z') \vee (y' \wedge z))') \vee (x' \wedge ((y \wedge z') \vee (y' \wedge z)))$$

$$\text{展开得 } (x \oplus y) \oplus z = x \oplus (y \oplus z)$$

故满足结合律

综上所述 $\langle B, \oplus \rangle$ 构成群

T₁) B 是布尔代数. $\forall a, b, c \in B$. 若 $a \leq c$

$$\text{则有 } a \vee (b \wedge c) = (a \vee b) \wedge c$$

证明布尔代数满足此性质

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

$$a \leq c \text{ 得 } a \vee c = c$$

$$\text{故 } a \vee (b \wedge c) = (a \vee b) \wedge c$$

证毕